WORKSHEET

Chapter 7: Trigonometric identities

1 Evaluate $\int_0^{\frac{\pi}{2}} \sin^2(x)dx$.

2 Evaluate $\int_0^{\frac{\pi}{4}} \tan^2(x)dx$.

3 Evaluate $\int_0^{\frac{\pi}{3}} \sin(x)\cos(x)dx$.

4 Evaluate $\int_0^{\frac{\pi}{12}} \cos^2(x)dx$.

5 Evaluate $\int_0^{\frac{\pi}{2}} \sin^2(2x)dx$.

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6 Evaluate $\int_0^{\frac{\pi}{8}} \tan^2(2x)dx$.

7 Evaluate $\int_0^{\frac{\pi}{4}} \sin(2x)\cos(2x)dx$.

8 Evaluate $\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \sin^2(x)\cos^2(x)dx$.

9 Show that $\int \sin(x)\cos(x)dx = -\frac{1}{4}\cos(2x) + c$.

10 Show that $\int 2\tan^2(x)dx = 2\tan(x) - 2x + c$.

11 Show that $\int \cos^4(x)dx = \frac{1}{32}\sin(4x) + \frac{1}{4}\sin(2x) + \frac{3x}{8} + c$.

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12 Show that $\int 2\sin(3x)\cos(3x)dx = -\frac{1}{6}\cos(6x) + c.$

13 Show that $\int 2\cos^2\left(\frac{x}{4}\right)dx = x + 2\sin\left(\frac{x}{2}\right) + c.$

14 Show that $\int 6\sin^2\left(\frac{x}{2}\right)\cos^2\left(\frac{x}{2}\right)dx = -\frac{1}{4}x + \frac{3}{64}\sin\left(\frac{16x}{3}\right) + c.$

15 Show that $\int \sec^2(2x)dx = \frac{1}{2}\tan(2x) + c.$

16 Show that $\int \cos^4(x)dx = \frac{1}{32}\sin(4x) + \frac{1}{4}\sin(2x) + c.$

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Answers

1 $\frac{\pi}{4}$

2 $\frac{3\pi}{4}$

3 $\frac{3}{8}$

4 $\frac{\pi}{24} + \frac{1}{8}$

5 $\frac{\pi}{4}$

6 $\frac{1}{2} - \frac{\pi}{8}$

7 $\frac{1}{4}$

8 $\frac{\pi - 2}{32}$

9–16 Proofs using relevant trigonometric identities.

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